

Retained Double-Commutator Bounds for a Wider Conditional Quantum Capture Window

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Abstract—We sharpen finite thermal-response curvature by retaining the mixed bath commutator after partial tracing. The bath-only outer commutator disappears, but its mixed contribution does not. In the source flat-band two-excitation reference, weighted exchange gives a uniform curvature ceiling $(2864/3)\nu$, with ν the finite occupation trace. This replaces the earlier $4 \cdot 64^2\nu$ charge. Combined with inherited derivative-launch certificates, response-point and slope bounds, Gaussian remainder and freshly extended uniform continuum value comparisons, rational arithmetic certifies retained error below $119/2^{23} < 2^{-16}$ throughout 16 ± 2^{-6} at unchanged $\beta = 5/16$. The interval halfwidth is fourfold larger than the preceding release. This is a conditional fixed-launch result, not permanent retention or hardware evidence.

Index Terms—Open quantum systems, finite reservoirs, trace bounds, quantum control.

I. CONTRACT AND SCOPE

Keep $\Omega = 64$, $B = 16$, $\kappa = 8$, $\beta = 5/16$, $T = 16$, $h = g = 1$ and $\lambda^2 = 4\sqrt{2}$. The launch is the source $p0$ state with the vacuum baseline. The thermal correction uses the $M = 192$ Gauss reference. Its finite identity [1] is

$$\Delta_{2,M}(t) = \text{Tr}_B X(t), \quad X(t) = U_2 \rho_{1,N} U_2^\dagger - \nu U_1 \rho_0 U_1^\dagger. \quad (1)$$

Both blocks embed in total charge $Q \leq 2$; $\nu = \text{tr } N$ and $\|X(t)\|_1 \leq 2\nu$. This finite occupation trace is not a uniformly bounded continuum photon count. The continuum comparison below transfers values, not derivatives.

Before evaluating this release we declared $w = 2^{-k}$, $k = 4, \dots, 24$, and horizon $U = 16 + 2^{-4} = 257/16$. This expands the preceding list and is post-exposure engineering, not an independent prediction. No temperature, state, target or physical resource is changed.

II. THE MIXED TERM MUST BE KEPT

Write $H = H_B + K$, $K = H_S + H_{SB}$. Bath-only commutators vanish under partial trace. Jacobi gives the exact retained identity

$$\text{Tr}_B[H, [H, X]] = \text{Tr}_B[K, [K, X]] \quad (2)$$

$$+ \text{Tr}_B[[K, H_B], X]. \quad (3)$$

Indeed, the outer H_B terms trace to zero, and $[K, [H_B, X]] = [[K, H_B], X] + [H_B, [K, X]]$. Discarding the mixed term would produce an unsupported curvature improvement. Differentiating $X(t)$ gives minus this retained double commutator.

MQM development v0.1, 7 October 2026. Same-agent derivation and separate controls. External review OPEN; physical promotion zero.

The inherited native-plus-exchange norm on $Q \leq 2$ is $\|K\| < 14$ [1]. To bound the remaining term, use the source lowering operator $S = I_C \otimes \sigma_R^-$:

$$H_{SB} = S^\dagger a(c) + S a^\dagger(c). \quad (4)$$

Commutation with the centered bath replaces c by the weighted vector δc , up to a phase. The same bright-mode exchange blocks therefore give

$$\|[K, H_B]\| \leq \sqrt{2} \|\delta c\|. \quad (5)$$

The flat-band moment, exact for Gauss degree at least two, is

$$\|\delta c\|^2 = G^2 B^2 / 3 = \frac{32768}{3\pi}. \quad (6)$$

Consequently $\|[K, H_B]\| < 256/3$, using $\pi > 3$. This is a frequency-weighted exchange bound, not a charge of the full bath spectral norm.

Theorem 1. *In the ideal finite $M = 192$ reference, for all times,*

$$\frac{1}{2} \|\Delta_{2,M}''(t)\|_1 \leq \frac{2864}{3} \nu. \quad (7)$$

Proof. Partial trace contracts trace norm on the Hermitian terms at issue. Double commutation by K costs at most $4\|K\|^2\|X\|_1$, while the mixed term costs at most $2\|[K, H_B]\|\|X\|_1$. Half their sum, with $\|X\|_1 \leq 2\nu$, is bounded by $\nu(4 \cdot 14^2 + 2 \cdot 256/3) = (2864/3)\nu$. $\square$

The inherited ideal finite slope bound is $\frac{1}{2} \|\Delta_{2,M}'(16)\|_1 < 2^{-20}$ [1]. With the directed source ceiling $\nu \leq \nu_U$, integration yields

$$\frac{1}{2} \|\Delta_{2,M}(t) - \Delta_{2,M}(16)\|_1 \leq 2^{-20} w + \frac{1432}{3} \nu_U w^2. \quad (8)$$

The coefficient is approximately 0.005122 instead of $1/8$. No new propagated 192-column checkpoint is required for this all-time ideal curvature bound. Numerical slope and point certificates retain their inherited arithmetic assumptions.

III. UNIFORM THERMAL LEDGER

The preceding derivative-launch packet [2] directly compared $p0$, $p1$ and $p0 + p2$ with the continuum native projection. Re-evaluate its normalized launch debit on the enlarged horizon; it is below 2.141×10^{-4} . The enclosed vacuum slope remains below 2^{-16} . Its local curvature bound $C_q(w)$ gives cold variation

$$d_q(w) = 2^{-16} w + C_q(w) w^2 / 2. \quad (9)$$

At $w = 2^{-6}$, $C_q(w)$ is approximately 0.003866. These decimals are diagnostics; exact returned entries and rational upper roots determine the replay.

The source centered comparison [3], now over $[0, U]$, gives response vacuum and covariance allowances below 2^{-29} and 2^{-55} respectively. These deliberately replace the smaller old-horizon slots 2^{-30} and 2^{-56} . They are uniform value comparisons, so numerical continuum derivative transfer remains OPEN. With $E_0 = 3/2^{18} + 2^{-128}$ and inherited x_w , the complete majorant is

$$E_\beta(t) < E_0 + d_q(w) + 3/2^{26} + 2^{-20}w \quad (10)$$

$$+ (1432/3)\nu_U w^2 + 2^{-29} + 2^{-55} \quad (11)$$

$$+ \frac{x_w^2}{2(1 - x_w)}. \quad (12)$$

Here $x_w = 12433696189517/58512905602490430 + 6V(16w + w^2/2)$, $V = 64/31457265$; the Gaussian remainder stays below 2^{-25} .

Theorem 2. *Under the inherited continuum, Gaussian and arithmetic assumptions, retained thermal error is below $119/2^{23} < 2^{-16}$ for $|t - 16| \leq 2^{-6}$.*

Proof. Exact rational evaluation gives a diagnostic total 1.348968×10^{-5} . Integer cross multiplication verifies the stated ceiling and target. Every term bounds the full closed interval. Candidate 2^{-5} fails this majorant, not the physical task. $\square$

IV. AUDIT AND REMAINING GATES

Six random-state tests on separately assembled $M = 4, 8$ occupation matrices verify the retained identity against the full generator. Maximum discrepancy is 5.890×10^{-15} . Two actual thermal curvature calculations compare dense double commutation with propagated dyad curvature, agreeing within 1.851×10^{-20} . Twenty-one candidate decisions receive separate integer checks. These same-agent controls are not independent external review or proof by finite sampling.

The halfwidth gain is 4 against 2^{-8} and 1024 against the original 2^{-16} interval. Relative halfwidth is 2^{-10} ; full width is 2^{-5} model time units. Model time, the population derivative ceiling and the channel target remain distinct typed quantities. Permanent retention, finite-bath replenishment, arbitrary-input diamond accuracy, calibrated device time and physical validation remain OPEN. The next useful tightening is the actual finite response curvature at T , with a third-order remainder and explicit continuum value comparison.

REFERENCES

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